

TikTok Claims Classification Project

Exploratory Data Analysis (EDA) - Executive Summary

ISSUE / PROBLEM

The TikTok data team aims to develop a machine learning model to classify user submissions as claims. In this phase of the project, the data must be analyzed, explored, cleaned, and structured before any model building can take place.

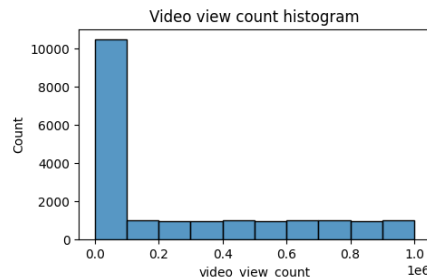
RESPONSE

At this stage, the TikTok data team conducted exploratory data analysis (EDA). The purpose of the EDA was to understand how videos influence TikTok users. To achieve this, the team analyzed variables reflecting user engagement, including view count, like count, and comment count.

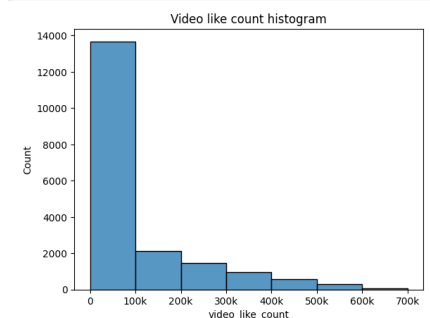
IMPACT

The exploratory data analysis indicated that the future claim classification model will need to address null values and the imbalance in opinion video counts. These factors should be considered when defining the model parameters.

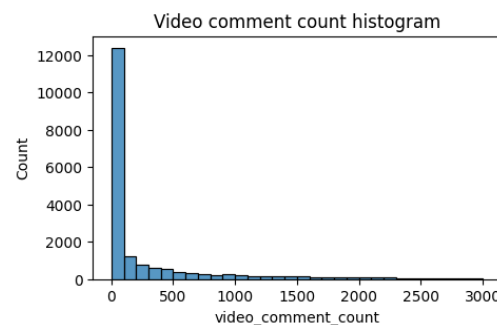
A key part of this project's exploratory data analysis is visualizing the data. As shown in the following histograms, the majority of videos are clustered at the lower end of the range for three variables that measure TikTok user engagement with the videos in this dataset.



The view count variable has a very uneven distribution, with more than half the videos receiving fewer than 100,000 views. Distribution of view counts > 100,000 views is uniform.



Similar to view count, there are far more videos with < 100,000 likes than there are videos with more.



Again, the vast majority of videos are grouped at the bottom of the range of values for video comment count. Most videos have fewer than 100 comments. The distribution is very right-skewed.

KEY INSIGHTS

The exploratory data analysis conducted by TikTok's data team highlighted several important considerations for the classification model, including missing values, the balance between claims and opinions, and the overall distribution of variables. Two key insights emerged from this analysis:

Null values

More than 200 null values were identified across seven columns. Future modeling should account for these missing values to avoid drawing conclusions based on incomplete data. Additional analysis is needed to understand the causes of the null values and their potential impact on statistical analysis and model building.

Skewed data distribution

Video view and like counts for opinion videos are concentrated at the lower end, around 1,000. As a result, the data distribution is right-skewed, which should be considered when selecting models and modeling approaches.